

Loan Prediction

Introduction

The dataset contains financial information about Indian individuals such as house and car ownership, annual income, Age, Marital status, occupation and living state. And we have a field, risk flag, that shows the person has defaulted on the loan or not.

Our goal is to find out which fields and properties can predict the Risk of default on a loan or even a check better and more effectively. And after that create a model to predict that take a person financial information and predict he/she might defaults on loan or not.

After downloading dataset from kaggle website we are going to visualize our data using Microsoft Power Bi to achieve a better understanding of data and its fields.

Using bar charts and cards we are going to show the effect of below factors on Risk flag our. Our measure for risk flag is 0 (hasn't defaulted) and 1 (has defaulted) which we gonna use the average of risk flag for each category in each factor and show the result. In all Dashboards you can filter the charts by clicking on bars.

Risk Flag, Average vs Sum

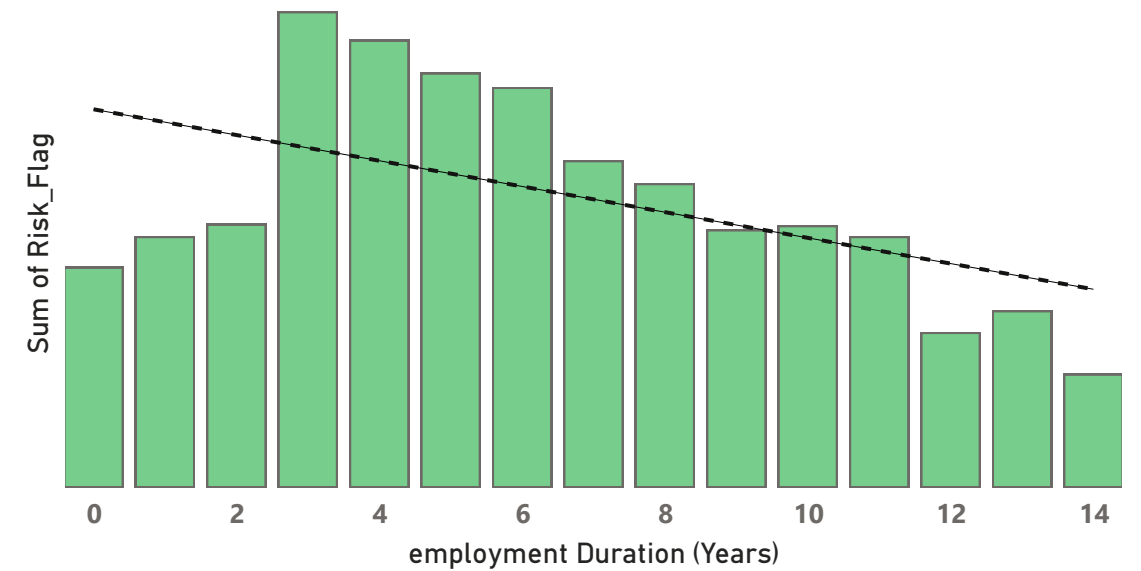
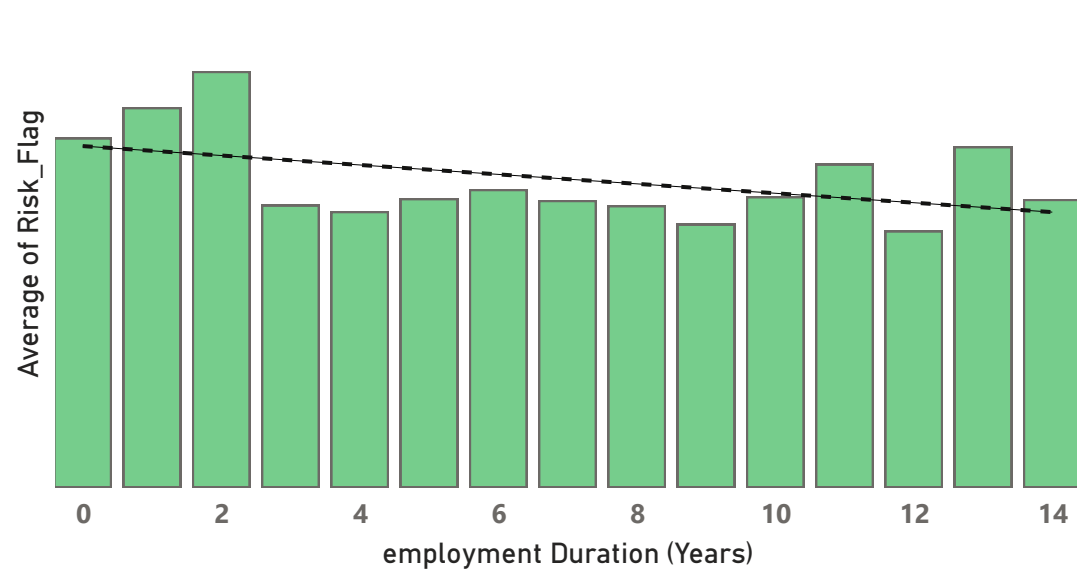
Risk flag in our visualization is default on loan which 0 means (did not default) and 1 means default in the dashboards you will see bar charts that shows the relation between different factors like profession and residence status and risk flag.

each factor has two chart one of them shows Average of Risk flag for each category and the other one shows sum of Risk flag for each category. Here we should determine the difference between Average concept and Sum concept.

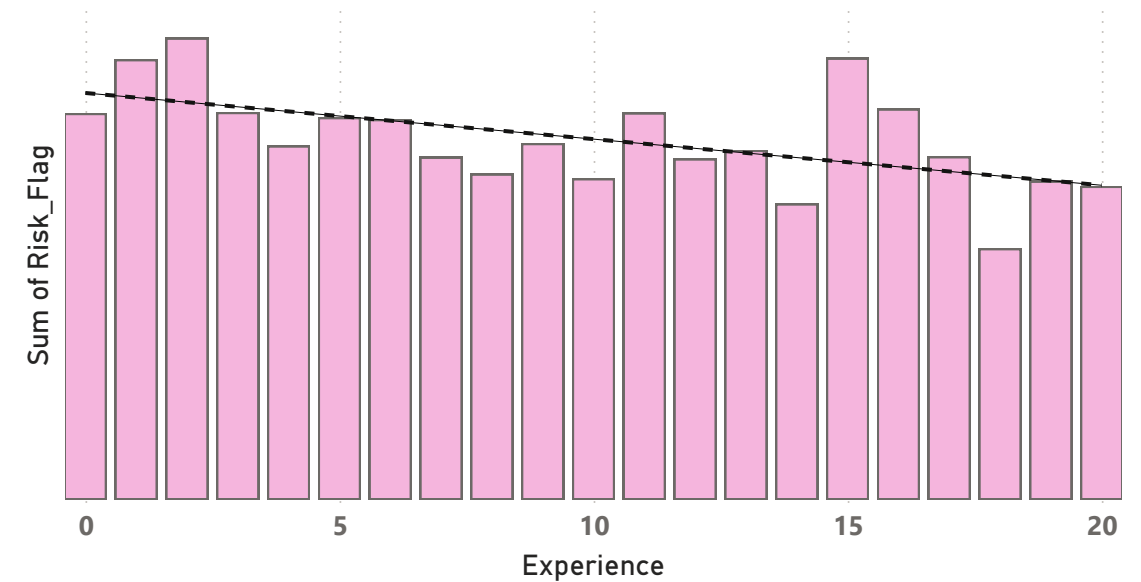
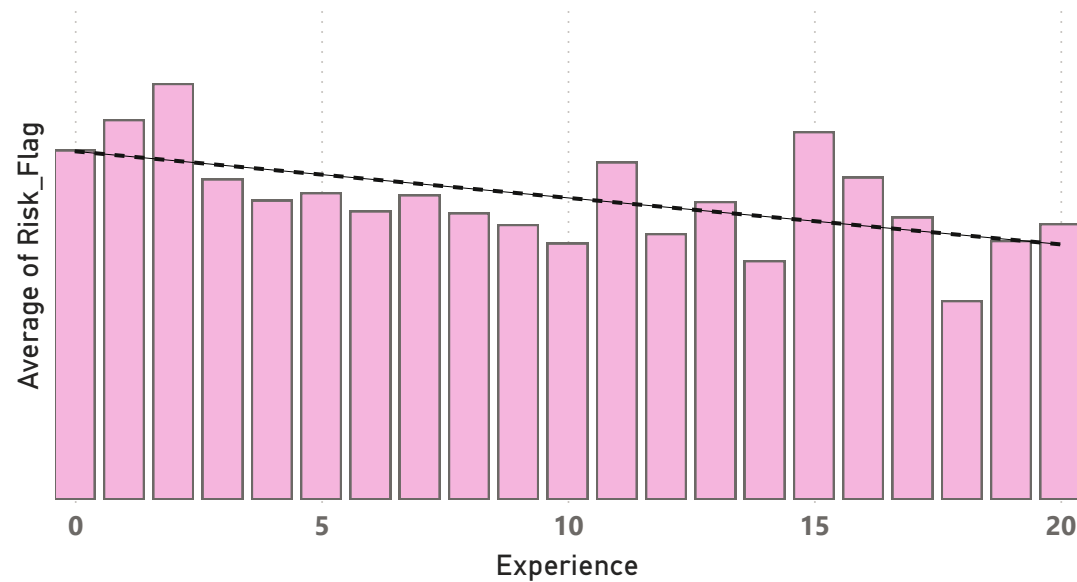
the Average shows Risk of a category with the consideration of Number of people that took loan for example if only 5 people of a category default on loan when the total count of people of the category who took loan was 10 the Average is 0.5 which is a high risk for that category but there might be a category that 1000 of them default on loan which is a big number but the total number of people who took loan is 10000 in this way the average is 0.1 which is low so to show the big number of 1000 people we use the Sum parameter which shows number of persons default on loan without consideration of count of all people who took loan.

in each dashboard, you will see two bar chart for each factor that one of the is Average Risk and the other one is the Sum (which is number of people who defaults)

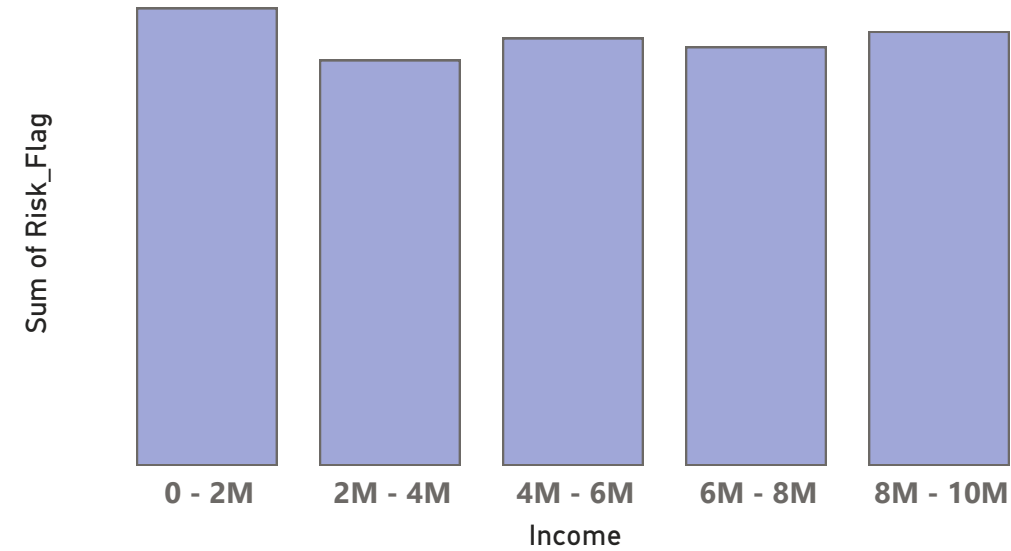
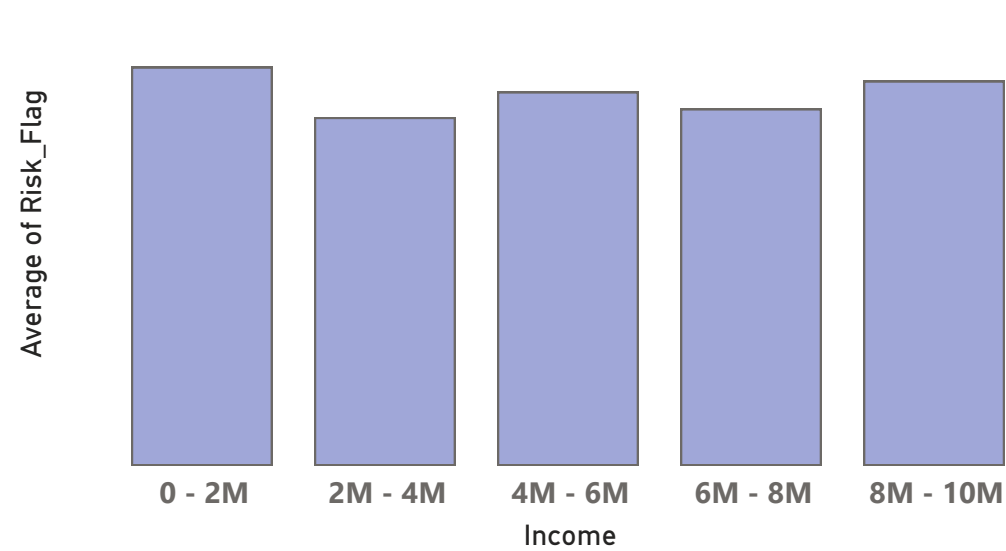
Employment Duration



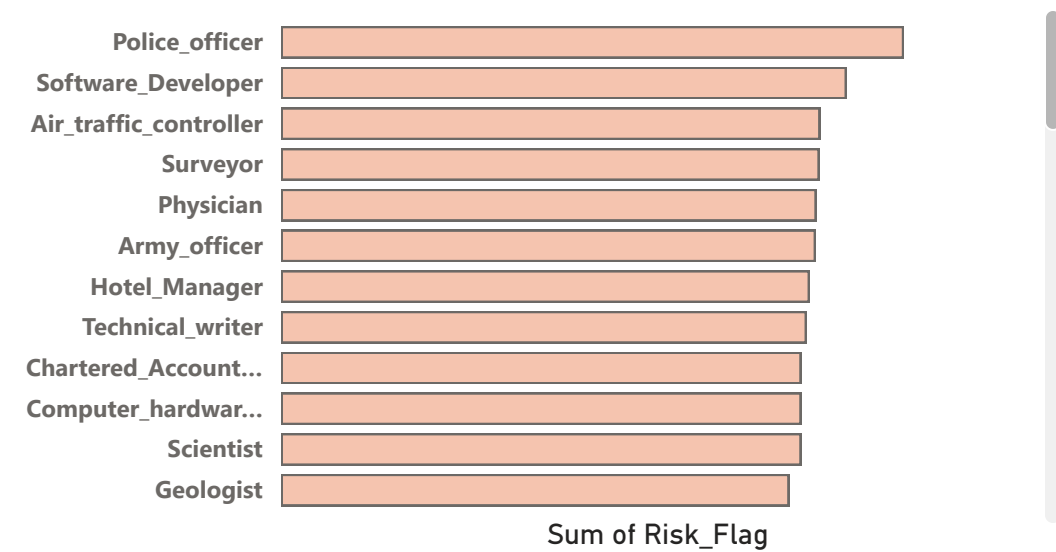
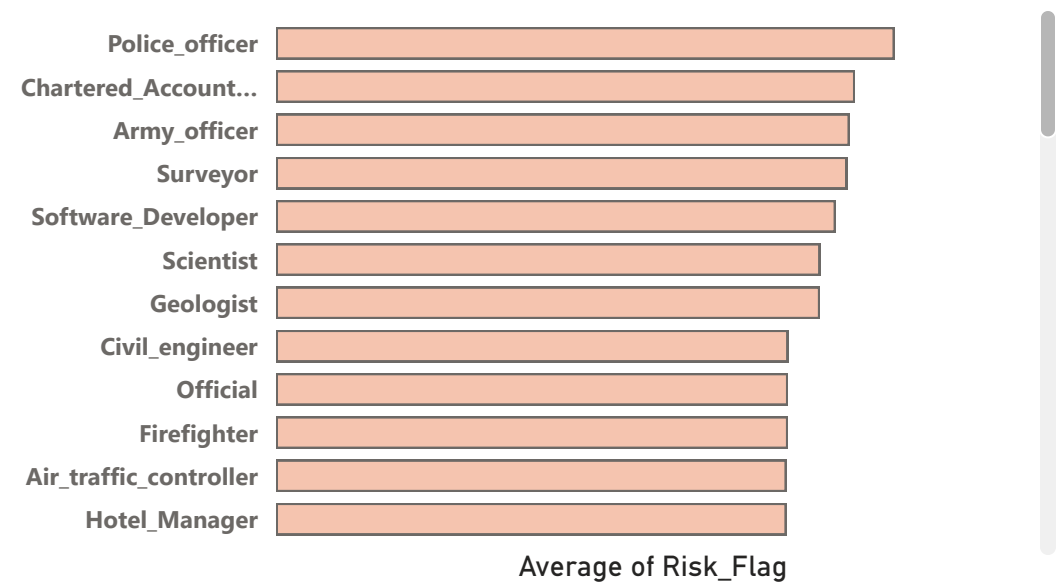
Experience



Income

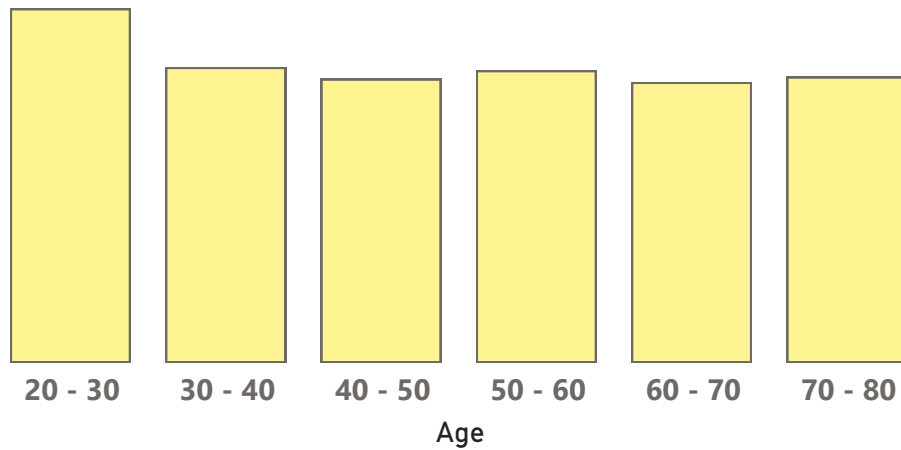


Profession

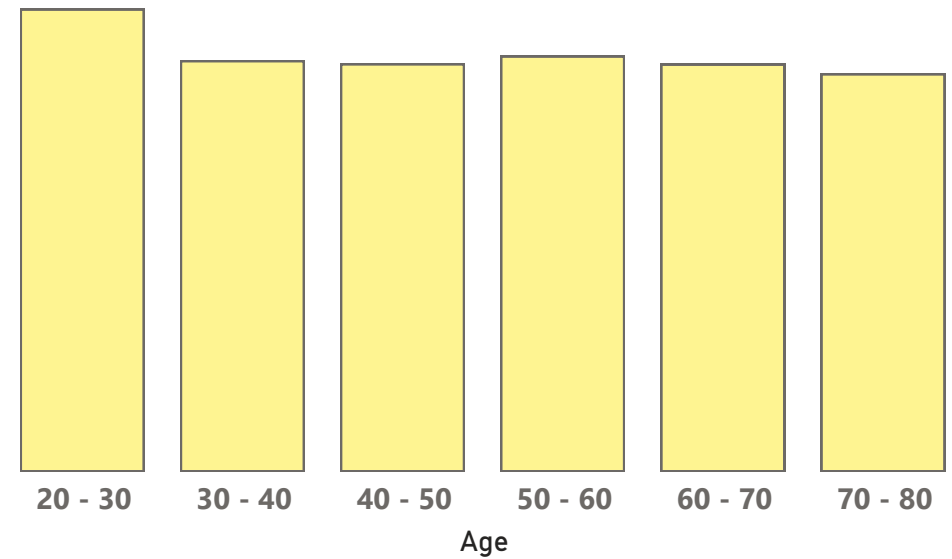


Age

Average of Risk_Flag

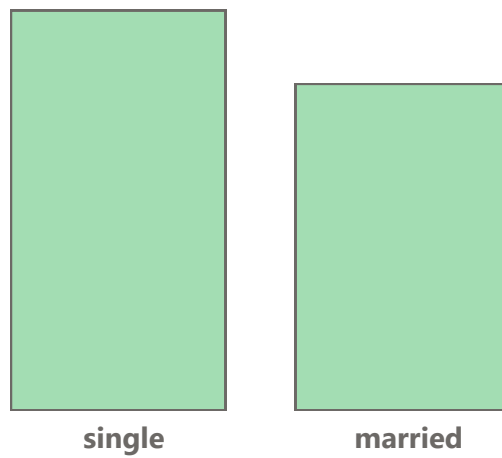


Sum of Risk_Flag

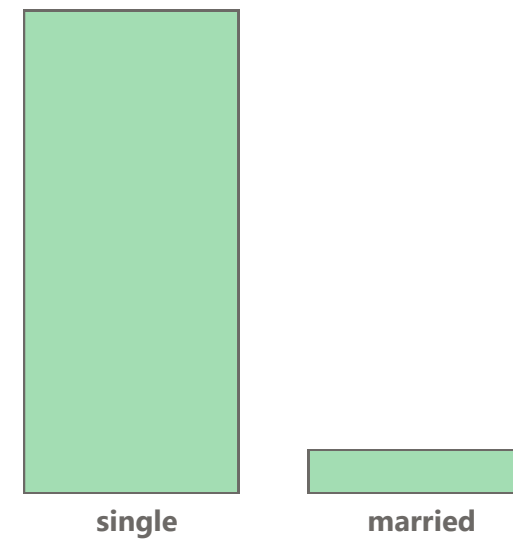


Marital Status

Average of Risk_Flag

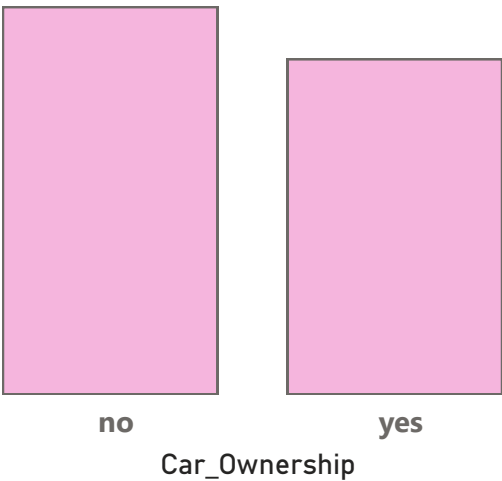


Sum of Risk_Flag

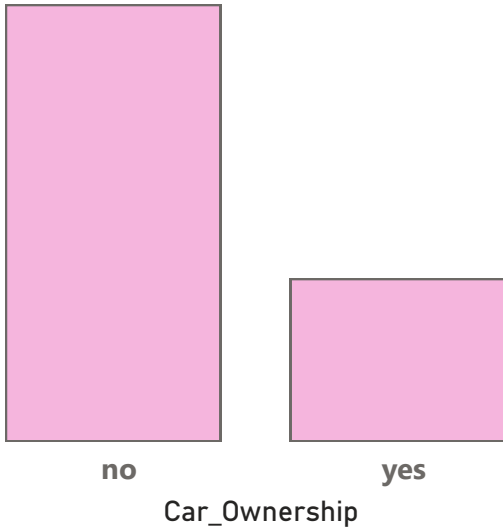


Car ownership

Average of Risk_Flag



Sum of Risk_Flag



House ownership

House_Ownership

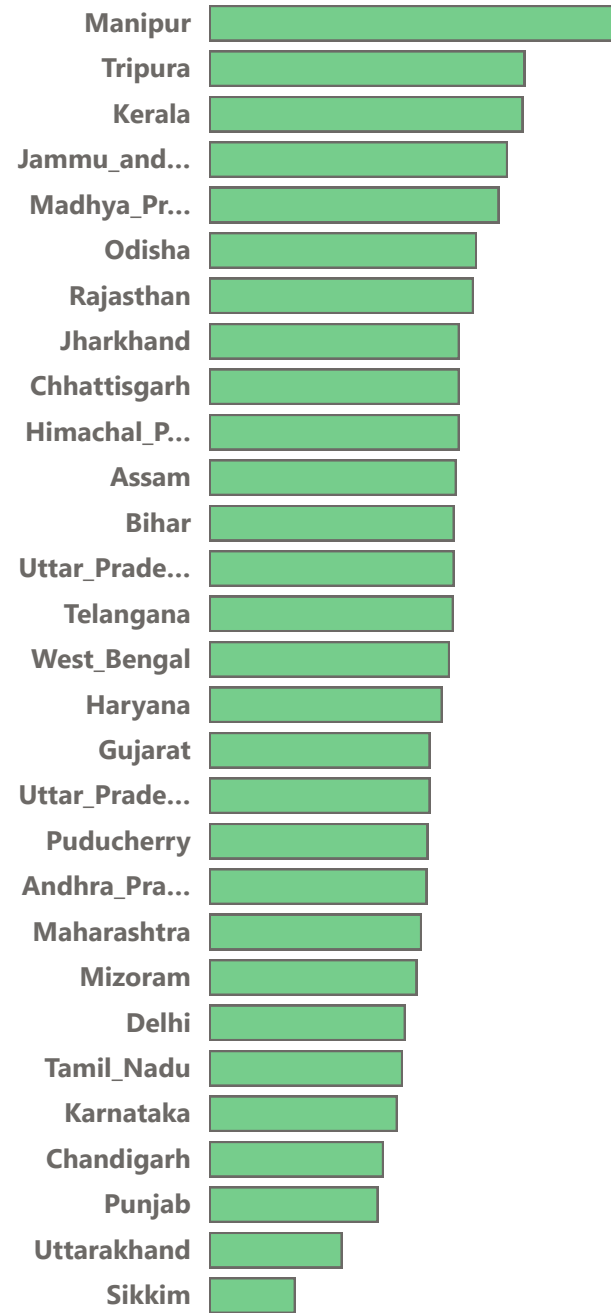
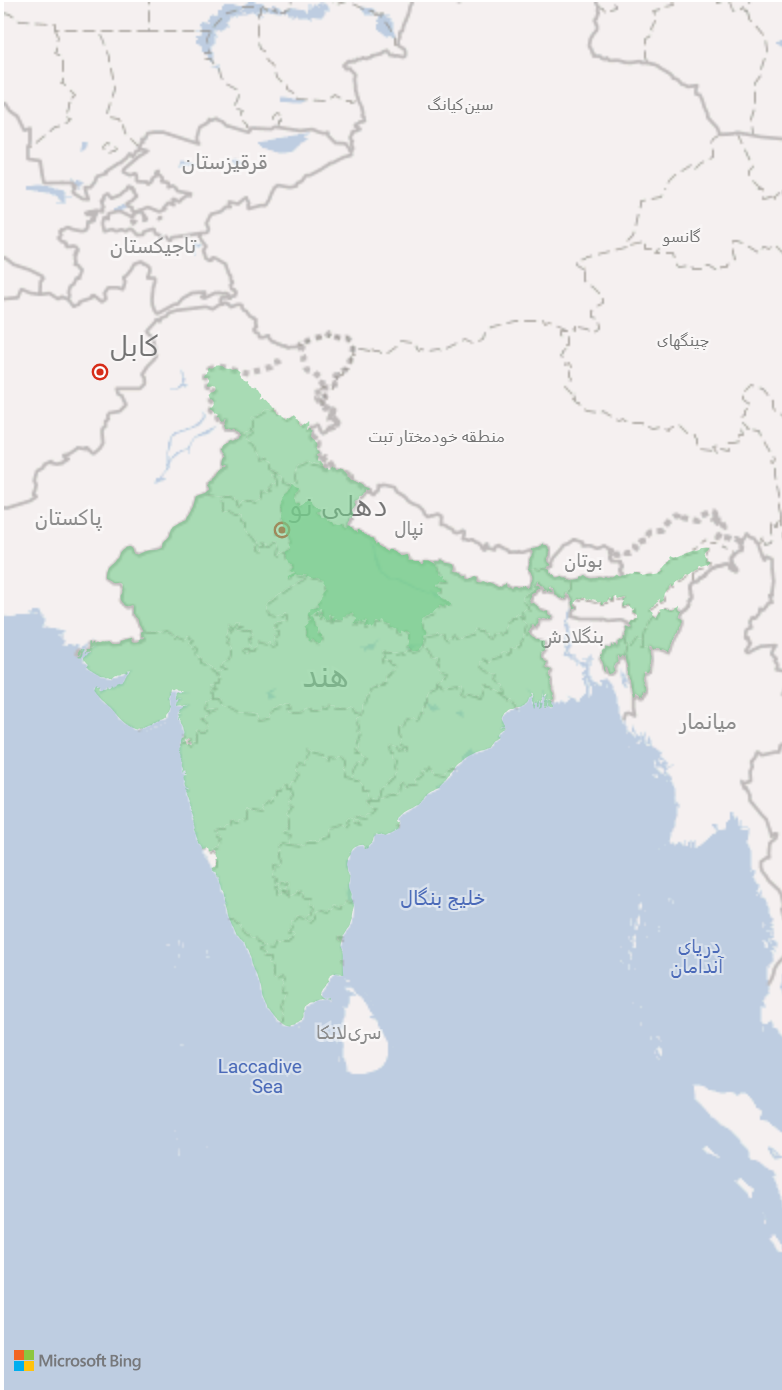


Average of Risk_Flag

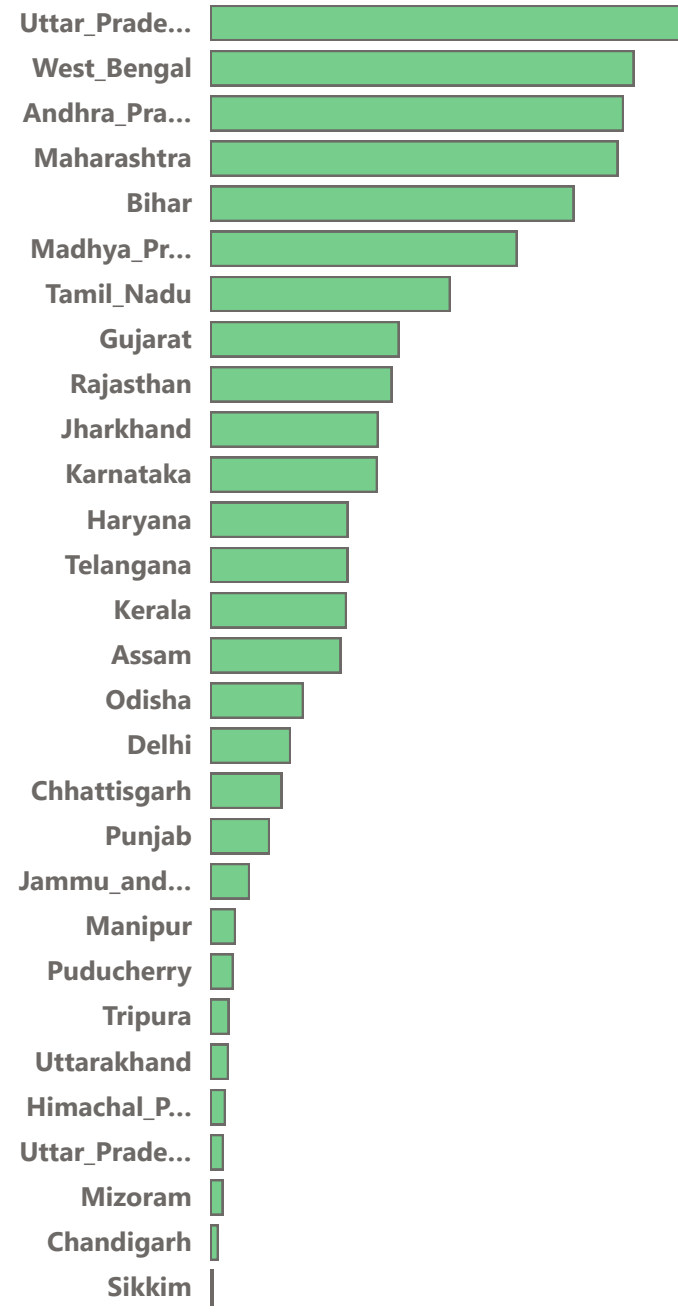
House_Ownership



Sum of Risk_Flag



Average of Risk_Flag



Sum of Risk_Flag